A Seminar Report on

Artificial Intelligence in Cybersecurity: A Review of IoT Security, Threat Detection and Smart Grid Defense Strategies

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by

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DECLARATION

The seminar work entitled "**ARTIFICIAL INTELLIGENCE IN CYBERSECURITY: A REVIEW OF IoT SECURITY, THREAT DETECTION AND SMART GRID DEFENSE STRATEGIES"** is carried out by me at JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, Kakinada under the guidance and supervision of Dr. N. Ramakrishnaiah, Professor & Head, Department of Computer Science and Engineering, UCEK, JNTUK, Kakinada.

This work is original and has not been submitted to any other University or Institute for the award of any degree or diploma.

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CERTIFICATE

This is to certify that the project titled **"ARTIFICIAL INTELLIGENCE IN CYBERSECURITY: A REVIEW OF IoT SECURITY, THREAT DETECTION AND SMART GRID DEFENSE STRATEGIES"** being submitted by **PULAPAKURA EMMI** bearing the Roll Number **25021D2607** in partial fulfilment for the award of the degree of **MASTER OF TECHNOLOGY** in **CYBER SECURITY** to UCEK, JNTUK, Kakinada, Andhra Pradesh, India is a record of bonafide work carried out under the guidance and supervision of the Department.

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Every piece of academic work carries with it the invisible contributions of people who guided, supported, and encouraged the person doing it. This report is no exception, and I want to acknowledge that honestly rather than in the usual formulaic way.

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I also thank the faculty and staff of the department for the academic environment they have maintained. Good academic culture is easy to take for granted until you have worked in settings where it is absent. I am grateful for it here.

To my family — who probably did not fully understand what this report was about but supported me through it anyway — thank you. And to my friends and fellow students who sat through the same late-night study sessions: this one is for all of us.

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ABSTRACT

Cybersecurity is no longer purely a matter of building better walls. The systems that need protecting — IoT-connected devices, industrial control networks, smart grid infrastructure, and AI-powered platforms — are too diverse, too interconnected, and too actively targeted for perimeter-based defences to remain adequate on their own. At the same time, the very artificial intelligence tools being adopted to strengthen defences are being weaponized by adversaries, creating an arms race that conventional rule-based approaches are ill-equipped to manage.

This report presents a comprehensive literature review of five peer-reviewed research papers published between 2024 and 2025, collectively examining how artificial intelligence is being applied across key cybersecurity domains. The selected studies span IoT cyber risk assessment, network security optimization, multi-sector threat detection, generative AI applications, and smart grid operational technology security — offering a broad yet focused cross-section of current AI-driven cybersecurity research.

Several core themes emerge consistently across the reviewed literature. First, machine learning and deep learning — particularly transformer-based architectures, federated learning frameworks, and unsupervised anomaly detection methods — provide capabilities that rule-based systems fundamentally cannot match: the ability to detect novel threats without prior signatures, to adapt as the threat landscape evolves, and to process data at scales no human analyst could manage. Second, explainability is increasingly recognized not as an optional enhancement but as a deployment prerequisite. AI security systems whose decisions cannot be interpreted and audited are unsuitable for regulated environments and high-stakes operational contexts, regardless of their raw detection accuracy. Third, generative AI introduces a dual-use challenge that the field has not yet adequately resolved: the same models that enable synthetic training data generation and automated threat analysis are equally available to attackers for crafting convincing phishing campaigns, generating deepfake social engineering content, and automating vulnerability discovery. Fourth, securing operational technology environments — particularly smart grids — requires approaches that differ fundamentally from conventional IT security, accounting for real-time constraints, legacy hardware, and the potential for cyberattacks to cause direct physical harm.

A recurring and critical limitation across all reviewed studies is the shortage of high-quality, real-world cybersecurity datasets, particularly for IoT and OT environments. Most AI models are validated on synthetic or controlled benchmarks, raising genuine questions about how well research results transfer to real-world deployments against adaptive adversaries. Governance and organizational factors are equally underrepresented — technical solutions alone are insufficient when patch management is absent, access controls are weak, or security culture is underdeveloped.

This report includes a cross-paper comparative analysis that identifies agreements, divergences, and gaps across the five studies, alongside a structured observations table documenting approach, findings, strengths, weaknesses, and proposed research directions for each paper. The conclusion reflects honestly on what these five studies collectively reveal: that AI-driven cybersecurity is advancing, necessary, and not yet keeping pace with the threats it needs to address.

Keywords: *Artificial Intelligence, Cybersecurity, IoT Security, Generative AI, Smart Grid, Threat Detection, Explainable AI, Federated Learning, Metaheuristic Algorithms, OT Security.*

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CHAPTER 1
INTRODUCTION

1.1 Background and Overview

There is a useful way to think about where cybersecurity currently stands: it is a field that has spent decades building increasingly sophisticated locks, only to find that the doors themselves have multiplied — and that many of the new doors were never designed to have locks at all. The Internet of Things has placed computing capabilities inside thermostats, pacemakers, industrial sensors, and traffic cameras. Smart grids have connected power infrastructure to internet-accessible control systems. And now, powerful AI tools are available to virtually anyone with an internet connection and the motivation to use them. Each of these developments is genuinely useful. Each also creates new vulnerabilities.

The traditional approach to cybersecurity — matching known attack signatures, applying static rules, blocking known-bad traffic — was never a permanent solution. It was always a way of defending against yesterday's attacks. The problem is that yesterday's attacks are not what organizations face today. Modern threats include zero-day exploits that target vulnerabilities no one has catalogued yet, polymorphic malware that changes its signature with every infection, AI-generated phishing emails that are indistinguishable from legitimate correspondence, and nation-state adversaries with the patience to conduct reconnaissance inside a target network for months before acting. Rule-based defences cannot keep pace with this. The question the research community has been trying to answer for the past several years is whether artificial intelligence can.

It is worth being clear about what that question actually involves. AI is not one technology — it is a collection of techniques, each with its own strengths, failure modes, and appropriate contexts. Machine learning models trained on historical attack data can identify patterns in new traffic that rule-based systems would miss, but they depend heavily on the quality and currency of their training data. Deep learning architectures can process raw network traffic or system logs at scales no human analyst could manage, but their decisions can be opaque and difficult to audit. Generative AI models can synthesize realistic attack scenarios for training purposes, but those same models can also generate convincing phishing emails on behalf of an attacker. Understanding what AI actually offers — and where its limits are — requires more precision than the broad claim that "AI is transforming cybersecurity" provides.

This report tries to bring that precision by examining five specific, peer-reviewed research contributions published in 2024 and 2025. They cover different domains — IoT systems, network security optimization, multi-sector threat detection, generative AI applications, and smart grid operational technology — but together they represent a reasonably comprehensive picture of where AI-driven cybersecurity research currently stands. Although all selected papers focus on AI-driven cybersecurity, they differ in their application domains including IoT systems, generative AI, threat detection, and smart grid infrastructure — and those differences matter as much as the common ground.

1.2 AI in Cybersecurity — Core Technologies

1.2.1 Machine Learning for Threat Detection

Supervised machine learning — where models learn from labelled examples of normal and malicious behaviour — has become the standard approach in many threat detection applications. Random forests, support vector machines, gradient boosting classifiers, and neural networks have all been applied with reasonable success to problems including network intrusion detection, malware classification, and spam filtering. Their advantage over rule-based approaches is real: a trained classifier can generalize from the patterns it has seen to identify new instances of similar threats, without needing an explicit rule for each case.

The limitation is equally real. Supervised models are only as good as their training data. If a new type of attack was not represented in the training set — a zero-day exploit, a novel malware variant, a previously unseen attack pattern — the model has nothing to generalize from. This is why unsupervised approaches have attracted growing interest. Anomaly detection methods, including isolation forests and autoencoders, do not require labelled examples of attacks; instead, they learn what normal looks like and flag anything that deviates significantly. The trade-off is a higher false positive rate, which creates its own problems when security analysts are already overwhelmed with alerts.

1.2.2 Deep Learning and Transformer Models

Deep learning — particularly Convolutional Neural Networks (CNNs) for spatial pattern recognition, LSTM networks for sequential data, and transformer-based architectures for long-range dependency modelling — has achieved state-of-the-art results on a range of cybersecurity benchmarks. Transformers, originally developed for natural language processing, have found an interesting role in cybersecurity because much of the data that security systems analyse is inherently sequential: log files,

network traffic, command histories, API call sequences. The self-attention mechanism in transformers is well-suited to identifying which elements in a long sequence are most relevant to each other.

Federated learning deserves separate mention because it addresses a structural problem rather than a purely technical one. Cybersecurity data is sensitive, legally constrained, and organizationally siloed. Hospitals cannot share patient network logs with other hospitals. Banks cannot share fraud transaction data with competitors. Federated learning allows ML models to be trained collaboratively — each participant updates model parameters on their local data and shares only those updates, not the underlying data. This makes it possible to benefit from large combined training sets without the privacy and legal risks of data centralization. The reviewed papers, particularly Dhanushkodi and Thejas (2024), identify federated learning as one of the more promising directions in multi-domain threat detection.

1.2.3 Explainable AI (XAI)

Explainability has moved from a research desideratum to a practical requirement. The reason is straightforward: when an AI system flags a network device as compromised or quarantines a server, a security analyst needs to know why. Without an interpretable explanation, the analyst either has to trust the black box — which undermines human oversight — or dismiss the alert — which defeats the purpose of having AI in the loop at all. Techniques like SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations) provide feature-level explanations for individual model predictions. They are not perfect tools — they are approximations, and in adversarial settings, attackers can potentially manipulate the features that the model identifies as important — but they are considerably better than no explanation at all.

Multiple papers in this review make XAI a central concern, which reflects a broader shift in the field. The research community has largely accepted that a model with 98% detection accuracy but no interpretability is less deployable than a model with 94% accuracy and meaningful explanations. For regulated industries and high-stakes security environments, this is not a philosophical position — it is a legal and operational necessity.

1.2.4 Generative AI — Both Sides of the Equation

Generative AI — large language models like ChatGPT, image generation systems like DALL-E, and related architectures — is the most significant recent development that this field has to grapple with.

On the defensive side, generative models can produce synthetic labelled attack datasets, help analysts interpret complex threat intelligence, and automate routine security documentation tasks. These are genuinely useful applications. On the offensive side, the same capabilities are already being used: AI-generated phishing emails that pass grammar checks and mimic legitimate organizational communication, voice cloning for phone-based social engineering, and automated tools that use language models to assist in vulnerability discovery. The Sai et al. (2024) paper reviewed in Chapter 2 engages with this dual-use reality directly, and it is one of the reasons that paper is worth reading carefully.

1.2.5 Metaheuristic Algorithms

Metaheuristic algorithms — genetic algorithms, particle swarm optimization, simulated annealing, and related approaches — are less prominently discussed in mainstream AI cybersecurity literature than neural networks or transformers, but they have real practical value in a specific class of problems: optimization under complex, non-differentiable constraints. Network configuration optimization — finding the routing, access control, and encryption settings for a mesh network that maximize security given hardware and performance constraints — is precisely this kind of problem. Nawaf and Bentotahewa (2025) make the case for metaheuristic approaches in this context, and while the paper has limitations in terms of empirical validation, the underlying argument is sound.

1.3 IoT, Generative AI, and Smart Grid Security

1.3.1 IoT Cybersecurity Challenges

The Internet of Things creates cybersecurity challenges that differ from those of conventional IT environments in several important ways. First, IoT devices are constrained: limited memory, limited processing power, limited battery life. The security controls that work on a corporate laptop — encrypted storage, running antivirus software, applying regular firmware updates — are often not feasible on a temperature sensor or a home security camera. Second, IoT deployments are heterogeneous: a single network might contain devices running dozens of different operating systems, firmware versions, and communication protocols. Maintaining visibility into this diversity is itself a non-trivial security challenge.

Third — and this is the insight that Radanliev et al. (2024) focus on — IoT creates interdependency risk. A single compromised device does not just represent a single security incident; it represents an entry point into a network of interconnected devices. The Mirai botnet, which exploited

default credentials on hundreds of thousands of IP cameras and DVRs to mount massive DDoS attacks, demonstrated this dynamic in 2016. The risk was not in any individual device; it was in their aggregate behaviour when compromised simultaneously. Any risk framework for IoT needs to model these interdependencies explicitly, which most existing frameworks do not.

1.3.2 Smart Grid and OT Cybersecurity

Smart grids sit at the intersection of two worlds that have historically been kept separate: Information Technology (IT), which handles data and business operations, and Operational Technology (OT), which controls physical processes like electricity generation and distribution. Connecting these two worlds enables significant operational benefits — real-time monitoring, automated demand response, integration of distributed energy resources — but it also removes the air gap that OT security traditionally relied on. Once a SCADA system is reachable from the internet, even indirectly through a connected IT network, it is reachable by attackers.

The 2015 attack on Ukrainian electricity distribution companies showed what this means in practice. Attackers with legitimate operator credentials — obtained months earlier through a spear-phishing campaign — remotely opened circuit breakers and cut power to approximately 225,000 customers. The attack was not technically sophisticated by contemporary standards; it relied on social engineering to obtain access and patience to learn the operational environment. The lesson is that OT cybersecurity is as much about access control, network segmentation, and monitoring as it is about any specific defensive technology. Lakhal et al. (2025) reviews the current state of AI and other technologies being applied to this problem, and it is arguably the most important paper in this review from a public safety standpoint.

1.4 Problem Statement

The papers reviewed in this report collectively point to a set of problems that remain genuinely unresolved. AI-powered security tools are improving, but they face persistent limitations: dependency on high-quality training data that is difficult to obtain for novel threats; opacity in deep learning models that makes their decisions hard to audit; adversarial vulnerabilities that allow attackers to deliberately manipulate model inputs to evade detection; and a generalization gap between performance on benchmark datasets and performance in real-world deployments.

Meanwhile, the offensive use of AI — particularly generative AI — is advancing at least as quickly as the defensive side. The same tools that security teams are starting to use for synthetic training data generation and automated threat analysis are available to attackers for phishing at scale and automated vulnerability discovery. There is no reason to assume that defenders will naturally stay ahead in this arms race; the history of cybersecurity suggests they often will not.

There is also a structural problem: cybersecurity research, policy, and operational practice operate in silos. IoT security research does not always connect with smart grid security research, even though smart grids increasingly contain IoT-class devices. Generative AI research does not always connect with threat detection research, even though detecting AI-generated attacks is precisely the problem that threat detection systems need to solve. And academic research does not always connect with the governance frameworks needed to deploy its outputs safely at scale. These gaps are not new, but they matter more as the underlying technology becomes more powerful and its deployment more consequential.

1.5 Objectives of This Study

This report has the following specific objectives:

- To review five peer-reviewed papers on AI and cybersecurity published in 2024–2025, covering IoT security, network optimization, multi-domain threat detection, generative AI, and smart grid OT security.
- To analyse each paper's methodology, findings, strengths, and weaknesses with genuine critical engagement rather than mere summarization.
- To conduct a cross-paper comparative analysis that identifies agreements, contradictions, and gaps across the five studies.
- To build an observations table that documents approach, findings, strengths, weaknesses, and proposed research directions for each paper.
- To draw conclusions that reflect honestly on what these papers reveal about the current state and near-term trajectory of AI-driven cybersecurity.

1.6 Scope and Limitations

This review covers five papers selected for their domain relevance and publication quality — all appear in internationally recognized venues including Frontiers in Big Data, Journal of Intelligent Systems, and IEEE Access. The review does not attempt to be comprehensive; five papers cannot cover every dimension of AI in cybersecurity. The selection prioritizes breadth across application domains

over depth in any single one. All five papers were published in 2024 or 2025, which means the review reflects a current snapshot of the field, though it will inevitably be partially outdated by the time of reading given the speed at which both AI capabilities and cyber threats evolve.

CHAPTER 2
LITERATURE REVIEW

2.1 Overview of Reviewed Studies

The five papers reviewed in this chapter were selected because, taken together, they offer a reasonably complete picture of where AI-driven cybersecurity research currently stands — its genuine contributions, its persistent limitations, and the gaps that future work needs to fill. They differ meaningfully in their methods: one develops an original risk model and validates it empirically, one synthesizes AI optimization approaches conceptually, one conducts a broad multi-domain literature review, one analyses specific generative AI models for dual-use potential, and one applies a rigorous systematic review protocol to smart grid security. They also differ in their application focus — IoT risk, network security, general threat detection, generative AI, and critical infrastructure — which means their findings are not always directly comparable but are often mutually illuminating.

All five are published in high-impact venues: Frontiers in Big Data, the Journal of Intelligent Systems (De Gruyter), and IEEE Access. The authorship spans institutions including the University of Oxford, University of Warwick, Vellore Institute of Technology, BITS Pilani, the National University of Singapore, and ENSAM Casablanca. This geographic and institutional diversity is worth noting — it reflects the genuinely global nature of both the AI cybersecurity research community and the threats it is trying to address.

Each paper is reviewed across four dimensions: research context and methodology, key findings, conclusion, and a critical assessment. The critical assessment is the part that matters most for this report's purposes; summarizing what papers say is less useful than evaluating whether what they say is convincing, complete, and applicable.

2.2 Detailed Review of Individual Papers

[1] AI Security and Cyber Risk in IoT Systems

Authors: *Petar Radanliev, David De Roure, Carsten Maple, Jason R. C. Nurse, Razvan Nicolescu, Uchenna Ani*
Journal: Frontiers in Big Data | Year: 2024 | DOI: 10.3389/fdata.2024.1402745

Research Context and Methodology

The starting observation of Radanliev et al. is one that practitioners in IoT security will recognise immediately: most existing cyber risk assessment frameworks were built for conventional IT environments and do not translate well to IoT. ISO 27005, NIST SP 800-30, and similar frameworks

assume relatively well-defined network perimeters, manageable device inventories, and endpoints capable of running security agents. IoT environments violate all of these assumptions. Devices may number in the millions, run stripped-down firmware that cannot be updated, and be physically deployed in locations where tampering is trivially easy.

The paper's response is a dependency-based risk model — one that explicitly captures the interdependencies among IoT components rather than treating each device as an independent risk unit. This is theoretically sound: in a densely connected IoT deployment, the risk profile of any single device depends heavily on what it is connected to, what data it handles, and what services it provides to other components in the network. The model integrates AI and machine learning for dynamic, real-time risk estimation, and incorporates XAI techniques — specifically SHAP-based feature importance — to keep the model's outputs interpretable to practitioners. Validation is conducted using the BoT-IoT dataset, a publicly available benchmark for IoT network intrusion detection. The paper also reviews emerging standards — SBOM, VEX, CSAF — for managing software supply chain risks in complex IoT software ecosystems.

Key Findings

The central finding is that interdependency modelling changes the risk picture significantly. A device that appears low-risk in isolation may be high-risk in context, if its compromise would provide an attacker with access to more sensitive components. The AI integration enables the model to adapt as the network state changes — new devices added, traffic patterns shifting — without requiring manual recalibration. XAI ensures that when the model flags a component as high-risk, there is an interpretable explanation available rather than just a score. The paper also makes a practical argument for cyber insurance as a risk transference mechanism, noting that the model's outputs can feed directly into insurance premium calculations.

Conclusion

The dependency model, empirically validated on BoT-IoT, provides a tool that can generalize across diverse IoT environments and contribute to a more secure and resilient IoT ecosystem. The authors position XAI and open datasets as essential components of trustworthy AI-driven IoT risk assessment, and they recommend SBOM and CSAF adoption as practical steps toward managing software supply chain risks.

Critical Assessment

This is the most methodologically original paper in the set of five, and the dependency modelling insight is genuine. However, a few limitations deserve honest acknowledgement. The BoT-IoT dataset, while useful, is synthetic — it was generated in a controlled research environment, not collected from real-world IoT deployments. How well the model performs on genuinely heterogeneous, real-world IoT data is an open question that the paper does not fully address. Second, the model's usefulness depends on having an accurate, up-to-date map of the dependency graph — which devices are connected to which, and what the downstream consequences of a compromise would be. In practice, many organizations lack this information. Large-scale IoT deployments often lack comprehensive, current asset inventories; building the dependency graph requires data that may simply not exist. This is acknowledged by the authors, but it is a more significant practical limitation than it receives credit for. Third, the cyber insurance component, while interesting, is underdeveloped. The actuarial models needed to price IoT cyber risk meaningfully do not yet exist at the scale the paper implies.

[2] Optimization of Cyber Security Through the Implementation of AI Technologies

Authors: *Liqaa Nawaf, Vibhushinie Bentotahewa*

Journal: Journal of Intelligent Systems | Year: 2025 | DOI: 10.1515/jisys-2024-0226

Research Context and Methodology

This paper takes a broader view than Radanliev et al., examining cybersecurity optimization through AI across multiple dimensions simultaneously: network security (specifically Wireless Mesh Networks), threat prediction via metaheuristic algorithms, data privacy through GDPR-compliant design, and organizational governance. The methodological approach is primarily conceptual and literature-based — the authors synthesize existing research on each of these topics rather than conducting original experiments. Metaheuristic algorithms (genetic algorithms, particle swarm optimization, and related methods) are positioned as the primary technical contribution: the argument is that WMN security configuration is an optimization problem that rule-based approaches and even standard gradient-based ML cannot adequately solve, but that metaheuristic methods can.

Key Findings

The paper finds that applying metaheuristic algorithms to WMN infrastructure improves network security resilience and provides more secure communications than conventionally configured networks. Neural networks that mimic human brain behaviour — the paper's phrasing, which is somewhat loosely

defined — are argued to be effective for predicting cybersecurity outcomes. GDPR's privacy-by-design and privacy-by-default principles are identified as applicable to IoT device design, addressing data subject privacy without requiring post-hoc compliance measures. Governance systems are identified as critical not just for enforcement but for communicating security principles across organizations. The paper also notes that COVID-19-driven digitalization accelerated cyber threat growth by dramatically expanding the attack surface of organizations that were not prepared for remote work security demands.

Conclusion

The authors conclude that effective cybersecurity requires combining network design, algorithm design, and human-computer interaction considerations — it is not purely a technical problem, and solutions that treat it as one will be incomplete. Metaheuristic algorithms, AI, and GDPR compliance together form a framework for cybersecurity optimization that addresses technical, regulatory, and organizational dimensions simultaneously.

Critical Assessment

The paper's genuine contribution is its insistence on treating cybersecurity as a sociotechnical problem rather than a purely technical one. The inclusion of governance, GDPR, and human factors alongside algorithmic approaches reflects a realistic understanding of how security works in organizations. However, the lack of empirical validation is a real limitation. The claims about metaheuristic algorithm performance are drawn from surveying existing studies rather than from original experiments, and the studies surveyed are not always directly comparable. It is not clear from this paper what performance improvements (in terms of attack detection rates, false positive rates, or other measurable metrics) metaheuristic optimization actually delivers over baseline configurations in WMN environments. The COVID-19 framing is accurate but feels somewhat dated in a 2025 paper — the pandemic is now several years behind us, and the specific security challenges it created have largely been absorbed into the broader remote work security literature. The GDPR-IoT integration, while conceptually sound, remains at the conceptual level without concrete implementation guidance.

[3] AI Enabled Threat Detection: Leveraging Artificial Intelligence for Advanced Security and Cyber Threat Mitigation

Authors: *Kavitha Dhanushkodi, S. Thejas*

Journal: IEEE Access | Year: 2024 | DOI: 10.1109/ACCESS.2024.3493957

Research Context and Methodology

If Radanliev et al. goes deep on a specific domain (IoT) and Nawaf and Bentotahewa take a broad conceptual view, Dhanushkodi and Thejas occupy the middle ground — a comprehensive literature review that covers AI-enabled threat detection across multiple sectors without developing an original technical contribution of its own. The sectors covered include network intrusion detection, IoT security, Industry 5.0, 5G networks, and autonomous vehicles. The paper surveys a wide range of AI techniques: supervised and unsupervised ML, deep learning architectures (CNN, LSTM, transformer models), federated learning, and blockchain-integrated detection frameworks. Throughout, the paper returns to two themes it treats as non-negotiable: explainability and resilience.

Key Findings

Transformer-based architectures outperform traditional sequential models (LSTMs, vanilla RNNs) on threat detection tasks involving long-range dependencies in log sequences and network traffic. Federated learning enables collaborative threat detection across distributed IoT environments without requiring data centralization, which is important both for privacy and for practical network topology reasons. Blockchain integration, when used to record model updates and detection events, improves the integrity and auditability of distributed detection systems. Explainability remains a significant gap: many high-performing models in the surveyed literature produce accurate outputs but cannot provide interpretable reasons, limiting their deployment in contexts that require human oversight. Real-time processing of high-volume network traffic and privacy preservation in federated systems are identified as persistent open challenges.

Conclusion

The review concludes that AI — particularly through ML and DL — is increasingly necessary for addressing modern cybersecurity challenges across the sectors examined. Substantial technical progress has been made, but the challenges of explainability, real-time processing, and privacy preservation have not yet been resolved. Federated and collaborative approaches represent the most promising direction for multi-domain, distributed security environments. Continued research and

cross-domain collaboration are needed to close the gap between research results and operational deployment.

Critical Assessment

This paper's strength is its breadth — it is the most comprehensive of the five in terms of domain coverage, and it will be useful to any reader wanting an orientation to the current landscape of AI threat detection research. Its weakness is also partly a function of that breadth: coverage across five domains in a single paper necessarily means shallow treatment of each. Specific techniques are introduced and assessed at a level of detail that is sufficient for orientation but not for application. The paper would benefit from quantitative benchmarking — specific precision, recall, and F1 scores for the techniques it discusses, compared on common datasets — rather than qualitative assessments alone. There is also a tendency to present the challenges (real-time processing, privacy, adversarial robustness) as problems awaiting solutions rather than as fundamental trade-offs. Federated learning, for instance, almost always reduces model accuracy relative to centralized training on the same data — the paper acknowledges this in passing but does not engage with the magnitude of the trade-off or the conditions under which it is or is not acceptable.

[4] Generative AI for Cyber Security: Analyzing the Potential of ChatGPT, DALL-E, and Other Models for Enhancing the Security Space

Authors: *Siva Sai, Utkarsh Yashvardhan, Vinay Chamola, Biplab Sikdar*
Journal: IEEE Access | Year: 2024 | DOI: 10.1109/ACCESS.2024.3385107

Research Context and Methodology

The Sai et al. paper has a specific methodological focus that distinguishes it from the other four: it analyses specific, named generative AI models — ChatGPT, DALL-E, and related LLMs — for their cybersecurity implications, both defensive and offensive. This dual-use framing is the paper's most important structural choice, and it is the right one. Any analysis of generative AI in cybersecurity that only considers the defensive side is incomplete in a way that will mislead practitioners. The methodology combines structured analysis of real-world applications and industry deployments (including those by Google, Microsoft, and other major technology companies) with evaluation of misuse scenarios.

Key Findings

On the defensive side: GAI models can generate diverse, realistic synthetic threat scenarios for training classifiers, addressing the chronic shortage of labelled attack data. ChatGPT and similar models can detect phishing attempts, summarize threat intelligence reports, and automate incident response documentation at speeds no human team could match. DALL-E and image generation models can create synthetic malware visualizations for CNN-based classification training. These are genuinely useful capabilities that several commercial security products are already beginning to incorporate. On the offensive side: AI-generated phishing emails are grammatically impeccable, contextually appropriate, and produced at essentially zero marginal cost per message — traditional linguistic and syntactic phishing filters are poorly equipped to detect them. Voice cloning enables phone-based social engineering impersonating known individuals. LLMs trained on security research can assist in generating proof-of-concept exploit code for known vulnerabilities, lowering the technical barrier to attack.

Conclusion

GAI holds significant potential for strengthening cybersecurity defences, but the same capabilities create serious offensive risks. Ethical guidelines, responsible use frameworks, and continuous advancement of AI security measures alongside GAI development are essential to prevent misuse. The paper calls for vigilance from both developers and users of GAI tools in security contexts.

Critical Assessment

This is perhaps the most practically relevant paper in the review — its subject matter is already affecting real security operations today, not in some notional future. The dual-use framing is handled responsibly, and the specific model-by-model analysis (ChatGPT for this, DALL-E for that) is more useful than generic claims about "generative AI" would be. Two limitations stand out. First, the field is moving very quickly, and specific model capabilities are already different from what they were when this paper was written in 2024. Some findings will age poorly. Second — and this is a substantive criticism — the paper's response to the dual-use problem is governance and ethics, which is the appropriate academic response but an incomplete practical one. Ethical guidelines do not prevent a threat actor who has already decided to act maliciously from using ChatGPT. The paper does not engage seriously with the technical and institutional mechanisms that would actually constrain misuse: model access controls, output monitoring, use-case restrictions. This gap between identifying the problem and providing actionable solutions is the paper's most significant weakness.

[5] Next-Generation Smart Grid Cybersecurity: A Systematic Review of OT Cyber Threats, AI-Driven Defense, Cyber Deception Techniques, and Emerging Security Strategies

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Research Context and Methodology

Lakhal et al. is the most methodologically rigorous of the five papers, applying the PRISMA 2020 systematic review protocol to the cybersecurity of smart grid OT systems. This is a domain where the stakes are unusually high: a successful cyberattack on electricity distribution infrastructure can cause physical power outages affecting millions of people, with safety consequences that go well beyond the financial losses associated with most data breaches. The paper's scope is comprehensive — it covers AI-driven anomaly detection, Edge Computing, Digital Twin technology, 5G/6G, Software-Defined Networking, cyber deception (honeypots and honeynets), and Blockchain — and the PRISMA methodology provides better guarantees against selection bias than the informal literature surveys used by Dhanushkodi and Thejas or the conceptual analysis of Nawaf and Bentotahewa.

Key Findings

AI anomaly detection significantly outperforms rule-based approaches for detecting False Data Injection attacks — a class of attack that manipulates sensor readings to deceive grid management systems into making incorrect control decisions. This is an important finding because FDI attacks are particularly dangerous precisely because they are designed to look legitimate. Edge Computing reduces detection latency by processing security-relevant data at network edges rather than centralizing it, which matters in OT environments where response time is measured in milliseconds. Digital Twin technology — maintaining a real-time virtual replica of the physical grid — enables two distinct security functions: continuous comparison between actual and expected behaviour (anomaly detection), and safe simulation of attack scenarios against the virtual grid without risking real operational disruption. Cyber deception techniques, including honeypots designed specifically for OT environments, are effective at detecting the reconnaissance activities that typically precede sophisticated attacks. Zero Trust Architecture is identified as a necessary paradigm shift for OT security — the traditional assumption that traffic inside the OT network is trustworthy is no longer tenable.

Conclusion

Smart grid modernization offers real benefits but substantially expands the attack surface compared to traditional isolated OT environments. Integrating AI, cyber deception, and emerging technologies like Digital Twins and SDN is necessary for smart grid resilience. Future research priorities

include developing real-world OT datasets, bridging the IT-OT security gap, exploring quantum-resistant and explainable AI approaches, and building international cooperation on smart grid security standards.

Critical Assessment

This is the strongest paper in the set by methodological standards, and its domain focus is both underexplored in mainstream cybersecurity literature and disproportionately important from a public safety perspective. The PRISMA methodology makes it more reproducible and less susceptible to cherry-picking than the other reviews in this set. The main limitation is inherent to the systematic review format in this domain: real-world OT cybersecurity datasets are scarce, which means most of the AI techniques reviewed are validated on simulated or synthetic data. How well these techniques perform against real attackers in real grid environments is an open empirical question. Additionally, while the paper covers many technologies, it does not provide clear guidance on how a utility with limited resources should prioritize them — should they start with AI anomaly detection, or cyber deception, or Zero Trust implementation? The paper's value is as a comprehensive technical map; a prioritization framework would make it more actionable.

2.3 Cross-Paper Comparative Analysis

Having reviewed each paper individually, it is worth stepping back to examine what they say collectively — where they agree, where they diverge, and what their combined picture looks like. This kind of cross-paper analysis is more revealing than reading the five papers in isolation, because it exposes both the points of convergence that the field has reached and the gaps that remain genuinely unresolved.

2.3.1 What the Papers Agree On

All five papers — explicitly or implicitly — share the view that rule-based and signature-based approaches are no longer sufficient for the threat environments their domains face. This is not a controversial position in the literature, but it is worth noting that the papers reach it from very different starting points: IoT risk modelling, network optimization, multi-sector threat detection, generative AI analysis, and OT security. The convergence from different directions gives the shared conclusion more weight than it would carry coming from a single perspective.

Explainability is a second point of convergence. Three of the five papers (Radanliev et al., Dhanushkodi and Thejas, and Lakhal et al.) make XAI a central concern rather than an afterthought. The

other two (Nawaf and Bentotahewa, Sai et al.) do not foreground XAI but do not contradict the argument either. This consistency suggests that the field has moved past debating whether explainability matters and is now focused on how to deliver it effectively. Dataset scarcity is the third shared theme: every paper, in some form, acknowledges that high-quality, real-world cybersecurity data — particularly for IoT and OT environments — is difficult to obtain, and that this limits the generalizability of research findings.

2.3.2 Where the Papers Diverge

The papers diverge most clearly on methodology. Radanliev et al. develop an original technical contribution and validate it empirically, which gives their findings the strongest evidentiary basis of the five. Lakhal et al. follow a rigorous systematic review protocol, which makes their findings more reproducible and less susceptible to selection bias than informal surveys. Dhanushkodi and Thejas conduct a broad but methodologically informal literature review — useful for orientation but less rigorous. Nawaf and Bentotahewa offer conceptual analysis without original experimental validation, which limits the confidence one can place in their specific performance claims. Sai et al. use structured analysis of real-world deployments, which is interesting but inherently anecdotal.

These methodological differences matter for how the findings should be weighted. Radanliev et al.'s claim that the dependency model improves risk assessment for IoT is supported by empirical evidence on a specific dataset. Nawaf and Bentotahewa's claim that metaheuristic algorithms outperform conventional approaches for WMN security optimization is supported mainly by surveying papers that support it — which is a weaker form of evidence. Readers who want to act on these findings should be aware of this variation.

2.3.3 Application Domain Differences and Their Implications

The five papers cover substantially different application domains, and this is not incidental — it reflects genuine specialization within AI cybersecurity research. IoT security (Radanliev et al.), general network optimization (Nawaf and Bentotahewa), multi-sector threat detection (Dhanushkodi and Thejas), generative AI (Sai et al.), and smart grid OT (Lakhal et al.) are all distinct enough that techniques that work in one domain do not automatically transfer to others. A dependency model designed for IoT risk assessment is not directly applicable to smart grid OT security, even though smart grids increasingly contain IoT-class devices. Generative AI techniques for synthetic training data

generation are relevant to the dataset scarcity problem identified in all domains, but this connection is not made explicit in any of the five papers.

This domain specialization is both a strength and a weakness of the current state of the field. It has allowed research to go deep on specific problems rather than remaining superficial across all of them. But it has also created silos — the insights from Radanliev et al. on interdependency risk could inform the smart grid security research of Lakhal et al., and the generative AI approaches in Sai et al. could partially address the dataset scarcity that Lakhal et al. identify as a limiting factor. These connections are implied but not explicit in the papers themselves, and the integration work remains largely undone.

2.3.4 Relative Strengths of the Five Papers

If the five papers are evaluated on methodological rigour, Lakhal et al. ranks first — the PRISMA systematic review protocol provides the most reliable and reproducible basis for the claims made. Radanliev et al. ranks second, as the only paper with original empirical validation of a novel technical contribution. Dhanushkodi and Thejas ranks third — the breadth of coverage is genuinely valuable even without original experimental contributions. Sai et al. ranks fourth — the dual-use analysis is important and the specific model focus is useful, but the governance recommendations are underdeveloped given the urgency of the problem identified. Nawaf and Bentotahewa ranks fifth — the governance and human factors perspective adds something the other papers lack, but the absence of empirical validation limits how much weight the technical claims can bear.

It is worth being clear that this ranking is about methodological rigour, not about practical importance. Sai et al.'s analysis of generative AI dual-use risk is arguably the most practically urgent contribution in the set, even if it ranks fourth on methodological rigour. And Nawaf and Bentotahewa's argument about governance being as important as algorithms is correct and undervalued in the field, even if the paper does not deliver the empirical evidence to fully support its technical claims. Ranking papers is a useful analytical exercise, but it should not substitute for reading them.

2.3.5 The Gap That No Paper Fully Addresses

Reading all five papers together, one gap stands out that none of them fully addresses: the adversarial arms race. Every AI-powered defensive capability described across these five papers can, in

principle, be studied and circumvented by a sufficiently motivated adversary. Adversarial machine learning — deliberately crafting inputs to defeat trained classifiers — is a documented and growing field. An attacker who knows that a network is protected by an isolation forest anomaly detector can, given enough knowledge of the model, craft traffic that the detector treats as normal. An attacker who knows that a grid's AI anomaly detection system is calibrated against a particular baseline can conduct an attack that stays within the normal range until the critical moment. These adversarial considerations appear briefly in Dhanushkodi and Thejas but are not engaged with seriously in any of the five papers. This is a significant gap, because the effectiveness of AI defences in adversarial settings — rather than in controlled research evaluations — is ultimately what matters.

2.4 Observations Table

Table 2.1 consolidates the key observations from all five reviewed papers. The proposed solutions column reflects this report's analytical contribution — each solution is intended to be specific and actionable, not merely aspirational.

Table 2.1: Observations of Literature Review – Approach, Findings, Strengths, Weaknesses & Proposed Solutions

S.No	Authors (Year)	Approach / Methodology	Key Findings	Strengths	Weaknesses	Proposed Solution
1	Radanliev et al.(2024)	Dependency-based IoT cyber risk model capturing component interdependencies; AI/ML for dynamic risk estimation; XAI (SHAP) for transparency; validated on BoT-IoT dataset; cyber insurance as risk transference; SBOM/VEX/CSAF standards review.	Interdependencies amplify IoT risk beyond individual device level; AI enables real-time adaptive assessment; XAI maintains interpretability; open datasets improve credibility; SBOM/VEX/CSAF enable supply chain risk tracking.	Only paper with original empirical validation; integrates XAI as a core component; covers emerging standards; strong multi-institution authorship (Oxford, Warwick, Kent, UCL, Keele).	BoT-IoT is synthetic — real-world generalizability unclear; model requires accurate asset inventory that many organizations lack; cyber insurance integration is underdeveloped.	Build a federated, lightweight AI risk framework with automated asset discovery and real-time SBOM compliance checking, validated on real heterogeneous IoT deployments.
2	Nawaf &Bentotahewa(2025)	Conceptual analysis of AI and metaheuristic algorithm optimization for WMN security; GDPR privacy-by-design integration into IoT; neural network-based threat prediction; governance and	Metaheuristic algorithms improve WMN security configuration; GDPR principles applicable to IoT device design; AI reduces cyber-attack frequency through better	Integrates technical, regulatory, and governance dimensions in one framework; highlights human-computer interaction as a security dimension; timely COVID-19 digitalization context.	No original empirical experiments; metaheuristic performance claims based on surveyed studies, not original data; GDPR-IoT integration remains conceptual; COVID-19 framing is dated for a 2025 paper.	Design a GDPR-compliant AI optimization engine for WMN security that benchmarks metaheuristic algorithm performance against standard IDS datasets under real network conditions.

S.No	Authors (Year)	Approach / Methodology	Key Findings	Strengths	Weaknesses	Proposed Solution
		human-computer interaction framing.	monitoring and prediction; governance is as important as technical controls.			
3	Dhanushkodi &Thejas(2024)	Comprehensive IEEE Access literature review of AI threat detection across IoT, 5G, Industry 5.0, autonomous vehicles; coverage of CNN, LSTM, transformer models, federated learning, blockchain integration; XAI and resilience as core requirements.	Transformers outperform LSTMs on sequential threat data; federated learning enables privacy-preserving collaboration; blockchain improves auditability; XAI is essential for deployment in regulated environments; real-time processing remains unsolved.	Widest domain coverage of the five papers; includes cutting-edge architectures; strong XAI and resilience emphasis; published in IEEE Access.	No original experimental results; federated learning accuracy trade-offs understated; lacks quantitative benchmarks across reviewed techniques; adversarial robustness not addressed seriously.	Develop a standardized multi-domain AI threat detection benchmark incorporating XAI scores, adversarial robustness metrics, latency requirements, and federated learning accuracy trade-off measurements.
4	Sai et al.(2024)	Structured dual-use analysis of specific GAI models (ChatGPT, DALL-E) for cybersecurity; maps defensive applications (synthetic training data, automated threat analysis) alongside offensive risks (AI phishing, deepfakes, exploit automation); includes industry deployment review.	GAI enables synthetic attack data generation addressing training data scarcity; automates threat detection and security documentation; but also enables AI-crafted phishing, deepfakes, and automated vulnerability exploitation at scale.	First specific GAI model-by-model analysis for cybersecurity; dual-use framing is honest and practically important; covers real industry deployments; BITS Pilani + NUS authorship.	Governance recommendations are conceptual and not enforceable; no empirical benchmarking of GAI vs. rule-based defences; deepfake detection countermeasures underdeveloped; findings will age quickly as models evolve.	Establish a Generative AI Responsible Use Framework for Cybersecurity with mandatory red-team testing, enforceable access controls, and technical attribution mechanisms for AI-generated attack content.
5	Lakhal et al.(2025)	PRISMA 2020 systematic review of smart grid OT cybersecurity; covers AI anomaly detection, Edge Computing, Digital Twin, 5G/6G, SDN, cyber deception (honeypots/honeynets), Blockchain, Zero Trust	AI anomaly detection outperforms rules for FDI attacks; Digital Twins enable safe attack simulation and continuous anomaly comparison; cyber deception detects early reconnaissance; Zero Trust is necessary for	Strongest methodology (PRISMA 2020); most comprehensive OT security coverage in the set; Digital Twins and cyber deception coverage fills real literature gap; published in IEEE Power & Energy Society Section.	Real-world OT datasets are scarce — most techniques validated on simulated data; no prioritization framework for utilities with limited resources; IT-OT integration gap acknowledged but unresolved.	Build an integrated smart grid security testbed combining Digital Twin simulation, AI anomaly detection, and OT-specific honeypots to generate realistic datasets and provide an evidence-based technology prioritization roadmap.

S.No	Authors (Year)	Approach / Methodology	Key Findings	Strengths	Weaknesses	Proposed Solution
		Architecture; literature from IEEE, ACM, Elsevier, Springer, MDPI.	OT; Edge Computing reduces detection latency.			

Reading the Table

Several patterns are worth drawing out from this table. First, the dataset scarcity problem appears in the weaknesses column of every single paper — sometimes directly (Lakhal et al.'s "real-world OT datasets are scarce"), sometimes indirectly (Radanliev et al.'s BoT-IoT validation, Nawaf and Bentotahewa's absence of original experimental data). This is not coincidence; it reflects a structural problem in the field that no individual paper can solve. What it suggests is that the highest-leverage research investment might be in shared data infrastructure rather than in new algorithms.

Second, the proposed solutions in the final column are not independent. A federated lightweight IoT risk framework (Paper 1's solution), a standardized AI threat detection benchmark (Paper 3's solution), and an integrated smart grid testbed (Paper 5's solution) all require similar underlying infrastructure: real-world datasets, federated learning pipelines, and standardized evaluation protocols. Building these components once, for shared use across research domains, would serve all three proposed solutions simultaneously and deliver more value than building them separately.

CHAPTER 3
CONCLUSION AND FUTURE WORK

3.1 Conclusion

This report set out to review five peer-reviewed papers on AI and cybersecurity, to compare and critically assess them, and to draw honest conclusions about what they collectively reveal. Having done that work, a few things stand out clearly.

The first is that AI is genuinely necessary in modern cybersecurity — not because it is fashionable, but because the alternative (rule-based defences against AI-augmented threats) is increasingly untenable. The papers make this case from different angles: Radanliev et al. show that IoT risk cannot be modelled without capturing interdependency dynamics that static frameworks miss; Dhanushkodi and Thejas demonstrate that transformer models and federated learning deliver detection capabilities that conventional ML cannot match at scale; Lakhal et al. find that AI anomaly detection is substantially more effective than rule-based approaches for detecting False Data Injection attacks in smart grids. The case for AI is not based on hype — it is based on specific, documented performance advantages in specific, documented contexts.

The second is that AI cybersecurity research still has a generalization problem. The impressive results reported in these papers — and in the broader literature they draw on — are mostly achieved on benchmark datasets, controlled research environments, or synthetic data. How well these results transfer to the messiness of real-world deployments, against adaptive adversaries who are specifically trying to defeat the detection systems, is a question that the papers raise but cannot fully answer. Radanliev et al. validate on BoT-IoT, which is synthetic. Lakhal et al. note that real-world OT datasets are scarce. Nawaf and Bentotahewa do not validate their metaheuristic claims at all. This is not a criticism of the individual papers — it reflects a genuine structural limitation of the field — but it is a gap that practitioners relying on research results need to be aware of.

The third observation, and perhaps the most important, is that the papers which engage most honestly with the dual-use and governance dimensions of AI cybersecurity — particularly Sai et al. on generative AI — are also the ones that reveal how incomplete the current response to those dimensions is. Calling for ethical guidelines and responsible use is the right thing to do in an academic paper. But ethical guidelines do not prevent a threat actor who has already decided to act maliciously from using

ChatGPT to craft phishing emails. The technical and institutional mechanisms that would actually constrain that — model access controls, output monitoring, enforceable use restrictions, international regulatory frameworks — are political and organizational challenges that the research community can inform but cannot solve by itself. It is worth being clear about that limitation.

Although all five selected papers focus on AI-driven cybersecurity, they differ in their application domains including IoT systems, generative AI, threat detection, and smart grid infrastructure — and those differences matter as much as the common ground. The field is not one thing; it is a collection of domain-specific research programmes that have identified the same broad problem (rule-based defences are insufficient) but are pursuing different solutions in different contexts. The integration work — connecting the IoT risk insights of Radanliev et al. with the smart grid security research of Lakhal et al., or connecting Sai et al.'s synthetic data generation capabilities with the dataset scarcity that limits all the other papers — remains largely undone. That integration is, arguably, the most valuable contribution the next wave of research could make.

On balance, reading these five papers is a useful exercise not because they resolve the questions they address, but because they make clearer exactly what those questions are and why they are hard. That is probably the best that a review paper can do, and several of these papers do it well.

3.2 Future Work

Based on the analysis in this report, the following research directions appear most valuable as next steps:

Federated, Lightweight AI Risk Assessment for IoT

The dependency model of Radanliev et al. needs to be extended to real-world, heterogeneous IoT environments and validated against real attack data rather than synthetic benchmarks. A federated approach — where risk models are trained locally on device data and only aggregated updates are shared — would address both the privacy constraints and the network topology limitations that prevent data centralization. Integrating SBOM and CSAF compliance checking directly into the risk assessment pipeline would also address the supply chain dimension that the paper identifies but does not fully operationalize. This is a tractable research programme that builds on genuinely solid theoretical foundations.

Standardized AI Threat Detection Benchmarks

The absence of standardized, multi-domain benchmarks for AI threat detection is a persistent impediment to progress. Without them, claims about model performance are difficult to compare and verify. A useful research contribution would be the development of a benchmark suite that includes: representative datasets from multiple domains (IoT, 5G, OT), standardized adversarial robustness tests, XAI quality metrics, and latency benchmarks for real-time processing requirements. This kind of infrastructure does not produce a single dramatic research result, but it enables better research across the field over time.

Generative AI Governance Frameworks With Technical Teeth

The gap identified in the Sai et al. analysis — between recognizing the dual-use problem and proposing enforceable solutions — needs to be addressed. Future work should develop technical mechanisms alongside governance frameworks: watermarking or attribution techniques that allow AI-generated content to be identified and traced; access control frameworks that allow GAI capabilities to be restricted to verified legitimate uses; and empirical red-team/blue-team studies that measure the actual effectiveness of AI-powered defences against AI-powered attacks under realistic conditions. The governance frameworks are necessary but not sufficient; they need to be grounded in technical mechanisms that make them enforceable.

Smart Grid Cybersecurity Testbed with Real OT Data

The most significant limitation of Lakhal et al., and of smart grid OT security research more broadly, is the absence of real-world attack datasets. A shared testbed — combining a Digital Twin replica of realistic grid infrastructure with AI anomaly detection and cyber deception honeypots — could generate realistic OT attack data in a controlled environment, providing the training and evaluation data that current research lacks. Such a testbed would also serve as a validation environment for the theoretical claims about Digital Twin and cyber deception effectiveness that the reviewed paper makes but cannot empirically support.

Cross-Domain Integration Research

The most underinvested research direction identified by this review is the integration of insights across domains. Specifically: applying the interdependency modelling approach of Radanliev et al. to smart grid environments (which contain many IoT-class devices), using the synthetic data generation capabilities identified in Sai et al. to address the dataset scarcity problems identified across all domains,

and developing unified security frameworks that span IT and OT environments rather than treating them as separate problems. This kind of integration work is harder to publish as a single paper and harder to fund as a single project, which probably explains why it has been underinvested. It is also likely to produce more practical impact than incremental improvements to domain-specific approaches.

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